

Day 3: Data Sources & Storage Modes

[DEF] Definition to Remember

Power BI connects to over 150+ data sources using three primary storage modes: Import Mode (in-memory VertiPaq compression), DirectQuery (live source pass-through), and Dual/Composite Mode (hybrid).

1. Common Data Sources

- * Flat Files: Excel (.xlsx), CSV (.csv), JSON, XML.

2. Storage Modes Compared

Feature	Import Mode	DirectQuery Mode	Composite / Dual Mode
Data Location	Loaded into memory (RAM) via VertiPaq	Remains at the underlying source (SQL)	Tables set to Import or DirectQuery as
Performance	**Blazing fast** columnar compression	Dependent on backend database speed	Optimized: High-speed aggregations with
DAX Capability	Full, unrestricted DAX function support	Limited DAX (must translate into native)	Full DAX on import tables; translated
Data Freshness	Requires scheduled refresh.	**Real-time** live data on every visual	Near real-time with scheduled aggregation
Dataset Limit	Up to 1 GB (Pro) or 400 GB (Premium).	No Power BI size limit (handles multi-terabyte)	Combines massive tables with cached data

[MUST-WRITE] Must-Master Skills & Interview Points

1. Why Import Mode is preferred in 90% of BI projects (VertiPaq speed and full DAX support).
2. When DirectQuery is mandatory: strict regulatory security, real-time data needs, or massive petabyte datasets.
3. How the Folder connector eliminates manual copy-pasting for recurring data drops.

[RECALL] Quick Recall

Import (Fastest, in-memory, scheduled refresh) vs DirectQuery (Live SQL pass-through, real-time, query limits)